FDE COURSE · HEALTHCARE CASE STUDY

Who should get a follow-up call when they leave the hospital?

Predicting 30-day readmission for patients with diabetes. 130 US hospitals, 1999–2008.

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1 · THE PROBLEM

Imagine 100 patients going home and time for only 10 follow-up calls.

WHO

A discharge planner

They have a fixed number of calls to give out.

WHEN

At discharge

Only what is known before the patient leaves can be used.

DECISION

Who gets a call?

Everyone here has diabetes. Who will come back?

The 100 and 10 are an example to make the problem concrete.

1 · THE PROBLEM

The question: will this patient be back within 30 days?

TOO BROAD

Ever readmitted

A return months later is not something a call this week could change.

OUR CHOICE

Within 30 days

The window the team can still change. The original study used it too.

DESCRIPTION ONLY

Three classes

None, late, early. Handy for describing the file, but a worklist is still yes or no.

UCI dataset 296 · Strack et al. 2014

2 · THE DATA

One row is one hospital stay.

101,766
stays in the file

71,518
different patients

130
US hospitals, 1999–2008

Every patient has diabetes. The file records the stay, the care given, and whether a later stay was seen.

UCI dataset 296 · Strack et al. 2014

2 · THE DATA

After leaving, how many patients came back?



Only about 1 in 9 stays ends in an early return.

“No readmission” means no later stay was seen in this hospital network. UCI dataset 296 · Strack et al. 2014

89%

Accuracy is the wrong score.

Predict that nobody comes back and you are right 89% of the time, yet you find no one who needs a call.

Better score: of the calls the team can make, how many reach people who really return within 30 days?

UCI dataset 296 · Strack et al. 2014

2 · THE DATA

Four catches in the data

Same person, many rows

101,766 stays come from 71,518 patients. Keep each patient in training or in testing, never both.

The dead cannot return

Remove patients who died or went to hospice before scoring anything.

Missing fields

Weight is 97% missing. Payer and specialty are each missing about half the time.

No dates, no hospital ID

So we cannot test across time or across sites, and care from 1999–2008 is not today's care.

UCI dataset 296 · Strack et al. 2014

3 · THE SOLUTION

Build it in four steps

1	**Lock the question**	Label: back within 30 days. User: the discharge planner. Moment: before the patient leaves.
2	**Clean the group**	Remove deaths and hospice. Keep one stay per patient, which leaves 69,984 stays.
3	**Write a simple rule**	Prior hospital visits, not sent home, age over 60. Score only as many patients as the team can call.
4	**Try one model**	Logistic regression. Keep it only if it beats the rule, and give each flagged patient a short reason.

Stop when the next step does not beat the previous one. UCI dataset 296 · Strack et al. 2014

TAKEAWAY

A decision we can defend comes first. A finished model comes later.

NEXT STEPS THIS WEEK

1

Confirm the outcome counts: 54,864 none, 35,545 late, 11,357 early.

2

Count patient IDs that appear more than once.

3

List the discharge codes for death and hospice, then drop those rows.

Limit: this file cannot support a live clinical model. It is from 1999–2008 and has no date or hospital ID.